

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:

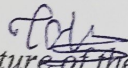
Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

(192512173)

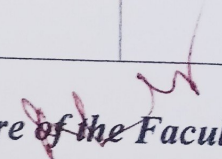
I Mohammed Taha M A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	2	2		
Accuracy	10%				
Clarity	5%				

Total Marks 28 /40


Signature of the Faculty

CO1-AT1

SHORT ANSWERS

6. Given,

Original file size = 900 MB

Compression = 40%

Encryption overhead = 5% of the compressed size

Compression reduces the file by 40%.

$$= \frac{900 \times (100 - 40)}{100}$$

$$= 900 \times \frac{60}{100}$$

$$= 540 \text{ MB}$$

Encryption overhead = 5% of 540 MB

$$= 540 \times \frac{5}{100}$$

$$= 27 \text{ MB}$$

Encryption file size = compressed size + Encryption overhead

$$= 540 + 27$$

$$= 567 \text{ MB}$$

Reduction in size = Original size - Encryption size

$$= 900 - 567$$

$$= 333$$

$$\text{Percentage reduction} = \frac{333}{9900} \times 100$$

$$= 3.37\%$$

Compression: Reduces file size, saves bandwidth, and speeds up transmission.

Encryption: Protects data by converting it into an unreadable format during transmission.

Decryption: Converts encrypted data back to its original form at the receiver.

7. Given,

$$\text{Total data} = 3 \text{ GB} = 3072 \text{ MB}$$

$$\text{Request size} = 3 \text{ MB}$$

$$\text{Throughput} = 120 \text{ Mbps}$$

$$\begin{aligned} \text{Total no. of requests} &= \frac{\text{Total data}}{\text{Request size}} \\ &= 3072 / 3 \\ &= 1024 \text{ requests} \end{aligned}$$

Total transmission time:

$$3072 \times 8 = 24,576 \text{ Mb}$$

$$= \frac{\text{Total data}}{\text{Throughput}}$$

$$= \frac{24,576}{120}$$

$$= 205 \text{ seconds}$$

Application layer services:

- Provides services such as FTP, HTTP, SMTP and DNS.
- Allows users to access network resources and transfer files.
- Ensures reliable communication by handling requests and responses.

works with lower layers to provide error free and efficient data transfer.

8. Given,

Application layer delay = 4 ms

Transport layer delay = 3 ms

Network layer delay = 5 ms

Data link layer = 2 ms

Physical layer delay = 1 ms

Propagation delay = 18 ms

Transmission delay = 12 ms

$$\begin{aligned}\text{Total processing delay} &= 4 + 3 + 5 + 2 + 1 \\ &= 15 \text{ ms}\end{aligned}$$

$$\begin{aligned}\text{Total end to end delay} &= 15 + 18 + 12 \\ &= 45 \text{ ms}\end{aligned}$$

Application Layer: Provide network services to user applications.

Transport layer: Ensures reliable data delivery and error recovery.

Network Layer: Determines the best route for packet delivery

Data link layer: Handles framing error detection and MAC addressing

Physical layer: Transmits data as electrical, optical or radio signals over the medium.

9. Given,

Useful data = 80 KIB

Frame size = 1600 Bytes

Protocol overhead per frame = 80 bytes

$$80 \text{ MB} = 80 \times 1024 \times 1024$$

$$= 83,886,080 \text{ bytes}$$

$$\text{No. of frames required} = \frac{83,886,080}{1600}$$

$$= 52,429 \text{ frames}$$

$$\text{Total overhead transmitted} = 52,429 \times 80$$

$$= 4,194,320 \text{ bytes}$$

$$\approx 4 \text{ MB}$$

$$\text{Transmission efficiency} = 1600 - 80$$

$$= 1520 \text{ bytes}$$

$$\text{Efficiency} = \frac{1520}{1600} \times 100$$

$$= 95\%$$

- Protocol overhead carries control information such as addresses and error checking data.
- It improves reliability and communication between devices.
- However, higher overhead reduces the amount of useful data in each frame.
- This increases bandwidth usage can reduce overall network performance